APPENDIX A

Elements of linear functional analysis

We briefly discuss basic terminology from linear functional analysis, which is the mathematical theory of linear operators on general vector spaces. All this can be taken from elementary textbooks on functional analysis.

A.1. Banach and Hilbert spaces

Definition A.1. Let E be a real vector space. A map $E \to \mathbb{R}$, $x \mapsto \|x\|$, is called a **norm on** E if

(i) $\|x\| \geq 0$ for all $x \in E$ and $\|x\| \geq 0$ iff $x = 0$;
(ii) $\|\alpha x\| = |\alpha|\|x\|$ for all $x \in E$ and $\alpha \in \mathbb{R}$;
(iii) $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in E$ (triangle inequality).

$(E, , \|\cdot\|)$ or simply E is called a **normed vector space**.

The following is obvious: Let $(E, , \|\cdot\|)$ be a normed vector space. Then

$$d(x, y) := \|x - y\|$$

for all $x, y \in E$ defines a metric on E.

Definition A.2. A normed space which is complete w.r.t. the metric induced by the norm is called a **Banach space**.

In other words: A normed vector space E is a Banach space iff every Cauchy-sequence $(x_n) \subset E$ is convergent (in E).

Definition A.3. Let E be a real vector space. A map $\langle\cdot,\cdot\rangle : E \times E \to \mathbb{R}$, is called a **scalar product on** E if

(i) $\langle x, y\rangle = \langle y, x\rangle$ for all $x, y \in E$;
(ii) $\langle x, x\rangle \geq 0$ for all $x \in E$ and $\langle x, x\rangle = 0$ iff $x = 0$;
(iii) $\langle \alpha x + \beta y, z\rangle = \alpha\langle x, z\rangle + \beta\langle y, z\rangle$ for all $x, y, z \in E$ and $\alpha, \beta \in \mathbb{R}$.

$(E, , \langle\cdot,\cdot\rangle)$ or simply E is called a **inner product vector space**.

Theorem A.4. *(Cauchy-Schwarz inequality) Let $(E, \langle\cdot,\cdot\rangle)$ be an inner product vector space. Then*

$$|\langle x, y\rangle| \leq \|x\|\|y\| \tag{A.1}$$

for all $x, y \in E$ with equality iff x and y are linearly dependent.

The following is obvious from (A.1): Let $(E, , \langle\cdot,\cdot\rangle)$ be an inner product vector space. Then

$$\|x\| := \sqrt{\langle x, x\rangle}$$

for all $x \in E$ defines a norm on E.

Definition A.5. (Hilbert space) An inner product vector space which is complete with respect to the induced norm is called a **Hilbert space**.

A.2. Orthogonality, orthonormal systems, Parseval's identity

Recall that two vectors u and v of an inner product space are called **orthogonal** if $\langle u, v\rangle = 0$.

Definition A.6. Let H be an inner product space with inner product $\langle\cdot,\cdot\rangle$ and induced norm $\|\cdot\|$. Let $M \subset H$ be a non-empty subset.

(i) M is called an **orthogonal system** if $\langle u, v\rangle = 0$ for all u,$v \in M$ with $u \neq v$.
(ii) M is called an **orthonormal system** if it is an orthogonal system and $\|u\| = 1$ for all $u \in M$.
(iii) M is called a **complete orthonormal system** or orthonormal basis of H if it is an orthogonal system and $\overline{\text{span}(M)} = H$.

From now on to the end of this section we assume that H is separable. Then any complete orthonormal system, in particular any orthonormal system, is finite or countable. Let $M = (x_i)_{i\in I}$ be an orthonormal system, then every $x \in \overline{\text{span}(M)}$ can be expanded in the following **abstract Fourier series**

$$x = \sum_{i\in I} \langle x, x_i\rangle x_i. \tag{A.2}$$

The series converges strongly in H. If $I = \mathbb{N}$, then

$$\lim_{N\to\infty} \|x - \sum_{i=1}^{N} \langle x, x_i\rangle x_i\|^2 = 0\,.$$

In particular, the **Pythagoras theorem** holds:

Theorem A.7.

$$\|x\|^2 = \sum_{i\in I} \langle x, x_i\rangle^2 \quad \forall x \in \overline{span(M)}. \tag{A.3}$$

Proof. If M is finite then

$$\begin{aligned}\|x\|^2 &= \|\sum_{i\in I} \langle x, x_i\rangle x_i\|^2 = \langle (\sum_{i\in I} \langle x, x_i\rangle x_i), (\sum_{i\in I} \langle x, x_i\rangle x_i)\rangle \\ &= \sum_{i,j\in I} \langle x, x_i\rangle\langle x, x_j\rangle\langle x_i, x_j\rangle = \sum_{i\in I} \langle x, x_i\rangle^2 \underbrace{\langle x_i, x_i\rangle}_{=\|x_i\|^2=1} \\ &= \sum_{i\in I} \|x_i\|^2\,.\end{aligned}$$

In the general case, w.l.o.g., assume that $I = \mathbb{N}$. Let $s_N := \sum_{i=1}^{N} \langle x, x_i\rangle x_i$. Then the finite case implies that

$$\|x\|^2 = \lim_{N\to\infty} \|s_N\|^2 = \lim_{N\to\infty} \sum_{i=1}^{N} \|x_i\|^2 = \sum_{i=1}^{\infty} \|x_i\|^2\,.$$

□

As a Corollary we obtain **Parseval's identity**.

Corollary A.8.

$$\langle x, y\rangle = \sum_{i\in I} \langle x, x_i\rangle\langle y, x_i\rangle \qquad \forall x, y \in \overline{span(M)}.$$

PROOF. $x, y \in \overline{\operatorname{span}(M)}$ implies $x + y \in \overline{\operatorname{span}(M)}$, hence

$$\begin{aligned}
2\langle x, y\rangle &= \|x+y\|^2 - \|x\|^2 - \|y\|^2 \\
&= \sum_{i\in I}\langle x+y, x_i\rangle^2 - \sum_{i\in I}\langle x, x_i\rangle^2 - \sum_{i\in I}\langle y, x_i\rangle^2 \\
&= \sum_{i\in I}\langle x, x_i\rangle^2 + 2\langle x, x_i\rangle\langle y, x_i\rangle + \langle y, x_i\rangle^2 - \sum_{i\in I}\langle x, x_i\rangle^2 - \sum_{i\in I}\langle y, x_i\rangle^2 \\
&= 2\sum_{i\in I}\langle x, x_i\rangle\langle y, x_i\rangle\,.
\end{aligned}$$

□

A.3. Linear operators

As already mentioned several times, a linear operator $L : E \to F$ between two Banach spaces need not be continuous if the domain E is infinite dimensional. A classical example is the derivative

$$Lf := \begin{cases} f' & \text{if } f \in C^1([0,1]) \\ 0 & \text{otherwise} \end{cases}$$

as a linear operator on the Banach space $C([0,1])$ of continuous functions on $[0,1]$ (endowed with the supremum-norm). Here, $C^1([0,1])$ denotes the subspace of one times continuously differentiable functions. Then, e.g. $f_n(x) := \frac{1}{n}x^n \to 0$ in $C([0,1])$, but $Lf_n(x) = x^{n-1}$ is not converging to 0.

In the following let E, F be two (real) normed vector spaces. We denote with $L(E,F)$ the space of **continuous** linear operators $L : E \to F$. We then have the following fundamental theorem of linear operators between normed vector spaces:

THEOREM A.9. *For $L : E \to \mathbb{F}$ linear, the following statements are equivalent:*

(i) *L is uniformly continuous;*
(ii) *$L \in L(E,F)$, i.e., L is continuous;*
(iii) *L is continuous at $x = 0$;*
(iv) *L is bounded;*
(v) *There exists $\alpha > 0$ such that $\|Lx\|_F \le \alpha\|x\|_E$ for all $x \in E$.*

PROOF. The implications $(i) \Rightarrow (ii)$ and $(ii) \Rightarrow (iii)$ are obvious. Suppose that L is continuous at $x = 0$ and let $U \subset E$ be any bounded subset, and let $M := \sup_{x\in M}\|x\|_E$ be its upper bound. We have to show that $L(U) \subset F$ is bounded. Since L is continuous at $x = 0$, we can find $\delta > 0$ such that $\|x\|_E \le \delta$ implies $\|Lx\|_F \le 1$. Consequently, if $x \in U$,

$$\|Lx\|_F = \frac{M}{\delta}\|L(\underbrace{\frac{\delta}{M}x}_{\|\cdot\|\le\delta})\| \le \frac{M}{\delta},$$

which shows that $L(U) \subset B^F_{\frac{M}{\delta}}(0)$. Suppose now that L is bounded, then there exists in particular $\alpha > 0$ such that $L(B^E_1(0)) \subset B^F_\alpha(0)$, and thus if $x \in E \setminus \{0\}$, then

$$\|Lx\|_F = \|x\|_E\|L(\underbrace{\frac{x}{\|x\|}}_{\|\cdot\|\le 1})\| \le \alpha\|x\|.$$

Since this is trivial if $x = 0$, we obtain (v). Finally, suppose that (v) holds. Then by linearity of L

$$\|Lx - Ly\|_F = \|L(x-y)\|_F \leq \alpha \|x-y\|_E,$$

showing the uniform continuity of L. □

Definition A.10 (operator norm). For $L \in L(E,F)$ we define

$$\|L\|_{L(E,F)} := \inf\{\alpha > 0 \mid \|Lx\|_F \leq \alpha \|x\|_E \text{ for all } x \in E\}.$$

We call $\|L\|_{L(E,F)}$ the **operator norm of** L.

The following Proposition summarizes all general equivalent characterizations of $\|L\|_{L(E,F)}$.

Proposition A.11. *Let $L \in L(E,F)$. Then $\|Lx\|_F \leq \|L\|_{L(E,F)} \|x\|_E$ for all $x \in E$. Moreover,*

$$\|L\|_{L(E,F)} = \sup_{x \in E \setminus \{0\}} \frac{\|Lx\|_F}{\|x\|_E} = \sup_{\|x\|_E = 1} \|Lx\|_F = \sup_{\|x\|_E \leq 1} \|Lx\|_F = \sup_{\|x\|_E < 1} \|Lx\|_F.$$

The space $L(E,F)$, endowed with the operator norm $\|\cdot\|_{L(E,F)}$, is again a normed vector space, and it is Banach space as soon as F (!) is a Banach space.

A.4. Duality

Definition A.12 (Dual space). If E is a (real) normed vector space, then

$$E' = L(E, \mathbb{R})$$

is the **dual space** of E. Elements of E' are often also referred to as **bounded linear functionals on** E. The operator norm on $L(E,\mathbb{R})$ is called the dual norm on E' and is denoted by $\|\cdot\|_{E'}$. If $f \in E'$, then we define

$$\langle f, x \rangle := f(x).$$

Remark A.13. By definition of the operator norm, the dual norm is given by

$$\|f\|_{E'} = \sup_{x \in E \setminus \{0\}} \frac{|\langle f, x \rangle|}{\|x\|_E} = \sup_{\|x\|_E \leq 1} |\langle f, x \rangle|$$

and as a consequence

$$|\langle f, x \rangle| \leq \|f\|_{E'} \|x\|_E.$$

A.5. Duality in Hilbert spaces

Suppose now that H is a (real) Hilbert space (with inner product $\langle \cdot, \cdot \rangle$). Since for all $y \in H$, the linear mapping

$$J(y) : H \to \mathbb{R}, x \mapsto \langle x, y \rangle$$

is continuous, we have that $J(y) \in H'$ with

$$\|J(y)\|_{H'} = \sup_{\|x\|_H \leq 1} |\underbrace{\langle J(y), x \rangle}_{= \langle x, y \rangle}| \leq \|y\|_H.$$

It follows that $y \mapsto J(y)$ is continuous, but in fact also the converse inequality holds, since we can choose the uni vector $x = \frac{y}{\|y\|_H}$ to conclude that

$$\|J(y)\|_{H'} = \sup_{\|x\|_H \leq 1} |\underbrace{\langle J(y), x\rangle}_{=\langle x,y\rangle}| \geq |\langle \frac{y}{\|y\|_H}, y\rangle| = \frac{\|y\|^2}{\|y\|} = \|y\|_H.$$

Consequently, $\|J(y)\|_{H'} = \|y\|_H$.

THEOREM A.14. *(Riesz representation theorem) The map $J : H \to H'$ is a linear isometric isomorphism. It is called the* ***Riesz isomorphism****.*

A.6. Reflexivity

The dual space $E'' := (E')'$ of E' is called the bi-dual space. Note that any $x \in E$ induces an element in the bi-dual space of E by considering the continuous linear functional

$$f \mapsto \langle f, x\rangle, f \in E'.$$

It follows that

$$\|x\|_{E''} = \sup_{\|f\|_{E'} \leq 1} |\langle f, x\rangle| \leq \sup_{\|f\|_{E'} \leq 1} \|f\|_{E'} \|x\|_E \leq \|x\|_E$$

hence the embedding $E \subset E''$ is continuous. In fact, due to the Hahn-Banach Theorem, there exists $f_0 \in E'$ such that $\|f_0\|_{E'} \leq 1$ and $\langle f_0, x\rangle = \|x\|$, which implies the converse inequality

$$\|x\|_E = |\langle f_0, x\rangle| \leq \sup_{\|f\|_{E'} \leq 1} |\langle f, x\rangle| = \|x\|_{E''},$$

so that

$$\|x\|_E = \|x\|_{E''}$$

and thus the embedding $E \subset E''$ is an isometry w.r.t. the respective norms. We say that a normed vector space E is **reflexive** if $E = E''$.

The most important class of reflexive spaces are Hilbert spaces (!), so in particular $L^2(\Omega, \mathcal{A}, \mu)$. Other important examples for reflexive function spaces are $L^p(\Omega, \mathcal{A}, \mu)$ for $p \in (1, \infty)$, since for $1 \leq p < \infty$ the space $L^{p^*}(\Omega, \mathcal{A}, \mu)$ with $p^* = \frac{p}{p-1}$ (and $p^* = \infty$ if $p = 1$) is the dual space of $L^p(\Omega, \mathcal{A}, \mu)$ and $p^{**} = p$ for $p \in (1, \infty)$.